

Evaluating Language Models

for Arabic E-Learning Excellence

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Abstract

This report presents a comprehensive comparative analysis of leading Large Language Models (LLMs) for Arabic e-learning applications, focusing on Modern Standard Arabic (MSA) capabilities. Through evaluation of performance on Arabic-specific benchmarks including ADMD, ArabicMMLU, and MGSM, we identify Claude 3.5 Sonnet and GPT-4/GPT-4o as the top contenders, with Claude achieving 30% accuracy on ADMD (highest) and 91.6% on MGSM, while GPT-4 leads with 72.5% on ArabicMMLU. The analysis covers core requirements, framework support, open-source alternatives, and provides strategic recommendations for platform integration.

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1 Overview of LLM Requirements for Arabic E-Learning

1.1 Core Capabilities

The Arabic e-learning platform requires LLMs with the following essential capabilities:

- **Content Generation:** Creating diverse, pedagogically sound learning materials in Modern Standard Arabic
- **Question Answering:** Providing accurate, context-aware responses across all difficulty levels
- **Conversation Practice:** Engaging in natural, fluent dialogues with constructive feedback
- **Grammar Correction:** Identifying and explaining errors with clear improvement suggestions

1.2 Language Focus

Modern Standard Arabic (MSA)

The platform requires LLMs with strong proficiency in MSA grammar, syntax, vocabulary, and stylistic conventions. Regional Arabic dialects are explicitly excluded from the requirements.

Key Requirements:

- Deep understanding of MSA grammar and syntax
- Contextually appropriate formal language generation
- Alignment with educational and professional standards
- Exclusion of colloquial dialect capabilities

1.3 Framework Support & Open-Source Options

1.3.1 LangChain/LangGraph Compatibility

LangChain¹ provides modular components for building LLM applications, managing prompts, connecting to data sources, and creating sophisticated question-answering capabilities. Most leading proprietary models offer API access with robust LangChain support.

1.3.2 Open-Source Alternatives

- **Jais 30B:** Bilingual Arabic-English model², trained on 126B Arabic tokens
- **Saba (Mistral):** 24B parameter model³ specialized for Arabic languages
- **AceGPT:** Open-source model specifically designed for Arabic educational applications

¹<https://www.linkedin.com/pulse/building-generative-ai-applications-langchain-sunil-kumar-u6ygc>

²<https://en.incarabia.com/core42-announces-opensource-arabic-llm-jais-30b-617603.html>

³<https://www.zdnet.com/article/mistrals-new-ai-model-specializes-in-arabic-and-related-languages/>

2 Benchmarking LLM Performance in Arabic

2.1 Arabic Depth Mini Dataset (ADMD)

ADMD⁴ features 490 challenging questions across 10 domains, manually evaluated for nuanced Arabic understanding and factual accuracy.

Claude 3.5 Sonnet achieved 30% accuracy - the highest performance, particularly strong in Mathematics (50% True) and Natural Sciences (45% True).

Table 1: ADMD Performance Comparison

Model	Accuracy (%)
Claude 3.5 Sonnet	30.0
GPT-4	11.8

2.2 ArabicMMLU Benchmark

ArabicMMLU⁵ contains over 14,000 multiple-choice questions from Arabic school exams, with >50% Arabic-specific contextual content.

GPT-4 achieved 72.5% accuracy - the highest score, surpassing Arabic-centric Jais-chat (62.3%) and demonstrating robust Arabic knowledge reasoning.

Table 2: ArabicMMLU Performance Comparison

Model	Accuracy (%)
GPT-4	72.5
Jais-chat (30B)	62.3

2.3 Multilingual Math (MGSM) Benchmark

The MGSM benchmark⁶ evaluates multilingual reasoning through grade-school math problems, indicating foundational Arabic comprehension capabilities.

Table 3: MGSM Top Performers

Model	Accuracy (%)
Claude 3.5 Sonnet	91.6
GPT-4o	90.5
Gemini 1.5 Pro	88.7

Strong MGSM performance indicates robust multilingual comprehension and logical reasoning abilities, essential for educational applications requiring step-by-step problem solving in Arabic.

⁴<https://arxiv.org/html/2506.01920v1>

⁵<https://aclanthology.org/2024.findings-acl.334.pdf>

⁶<https://www.vellum.ai/blog/llm-benchmarks-overview-limits-and-model-comparison>

3 Comparative Performance of Leading LLMs

Table 4: Comprehensive LLM Performance Comparison

Model	MGSM (%)	Ara-bicMMLU (%)	ADMDQ&A (%)		Key Strengths	Considerations
Claude 3.5 Sonnet	91.6	N/A	30.0	N/A	Leading MGSM & ADMD, strong reasoning, nuanced Arabic	Top contender. Excellent LangChain support via API
GPT-4 / GPT-4o	90.5	72.5	11.8	51.18	Leading ArabicMMLU & Q&A, strong multilingual	Robust API & LangChain support. Potentially high cost
Gemini 1.5 Pro	88.7	N/A	Mod-erate	10.83	Strong MGSM, good multilingual, large context	Variable Arabic task performance. Good LangChain support
DeepSeek V3	79.8	N/A	N/A	N/A	Good MGSM, strong reasoning, open-source	Limited Arabic benchmark data. Cost-effective option
Jais (30B Chat)	N/A	62.3	N/A	N/A	Open-source Arabic-centric, good ArabicMMLU	Specialized for Arabic. Fine-tuning potential
AceGPT (13B Chat)	N/A	~50.81	N/A	23.78	Arabic-centric, fine-tuning base, competitive	Open-source, specialized for Arabic education

3.1 Claude 3.5 Sonnet Analysis

Top Performer in Nuanced Arabic Understanding

Demonstrated exceptional performance on ADMD (30% accuracy) and leading MGSM score (91.6%), indicating superior capabilities in handling complex Arabic queries requiring cultural understanding.

Strengths: Strong reasoning, cultural nuance comprehension, balanced performance across domains

Key Metrics: 50% True in Mathematics, 45% True in Natural Sciences

Integration: API access with LangChain/LangGraph support

3.2 GPT-4 / GPT-4o Analysis

Leader in Arabic Knowledge & Reasoning

Achieved highest ArabicMMLU score (72.5%) and strong performance in specialized Arabic Q&A (51.18%), demonstrating robust general knowledge and reasoning in Arabic contexts.

Strengths: Broad Arabic knowledge, strong multilingual capabilities, proven educational applications

Key Metrics: 72.5% ArabicMMLU, 90.5% MGSM, 51.18% specialized Q&A

Integration: Mature API ecosystem with extensive LangChain support

4 Key Findings from Arabic-Specific Benchmarks

4.1 Claude’s ADMD Excellence

Claude 3.5 Sonnet’s 30% accuracy⁷ on ADMD significantly outperformed GPT-4’s 11.8%, particularly in challenging domains like Islamic Studies where other models exceeded 80% false rates.

Implication: Superior handling of nuanced Arabic content requiring cultural understanding.

4.2 GPT-4’s ArabicMMLU Dominance

GPT-4’s 72.5% score⁸ on ArabicMMLU surpassed Arabic-centric models like Jais-chat (62.3%), demonstrating comprehensive Arabic knowledge across 40 academic tasks.

Implication: Exceptional broad Arabic knowledge and reasoning capabilities.

4.3 Benchmark Limitations

Current benchmarks have limitations: ADMD’s small size (490 questions), ArabicMMLU’s uneven regional representation, and exclusive focus on MSA excluding dialects.

Recommendation: Multi-faceted evaluation approach with custom assessments.

4.4 ADMD Performance Breakdown

4.4.1 Claude 3.5 Sonnet Domain Performance

Table 5: Claude 3.5 Sonnet ADMD Domain Performance

Domain	True (%)
Mathematics & Computational Sciences	50
Philosophy & Logic	50
Natural Sciences	45

5 Synthesis of LLM Suitability for Arabic E-Learning

5.1 Top Contenders Based on Arabic Task Performance

Based on comprehensive benchmark analysis, the following models emerge as leading candidates:

Claude 3.5 Sonnet

- **91.6% MGSM** - Leading multilingual reasoning
- **30% ADMD** - Best nuanced Arabic understanding
- Superior cultural context comprehension
- Strong LangChain integration via API

GPT-4 / GPT-4o

- **72.5% ArabicMMLU** - Superior Arabic knowledge
- **51.18% Q&A** - Strong specialized task performance
- Robust general multilingual capabilities
- Mature API ecosystem and developer tools

⁷<https://arxiv.org/html/2506.01920v1>

⁸<https://aclanthology.org/2024.findings-acl.334.pdf>

5.2 MSA Support & Nuanced Understanding

Both Claude 3.5 Sonnet and GPT-4 demonstrate strong foundational understanding of Modern Standard Arabic, going beyond simple translation to grasp Arabic grammar, syntax, morphology, and cultural context.

Key capabilities:

- Grammar correction accuracy
- Stylistic appropriateness
- Cultural context understanding

5.3 Framework Support & Open-Source Options

5.3.1 LangChain Integration

Both leading models offer robust API access with comprehensive LangChain support⁹, enabling sophisticated application development and integration workflows.

5.3.2 Open-Source Alternatives

- **Jais 30B**: Specialized Arabic-English model¹⁰
- **AceGPT**: Education-focused Arabic LLM
- **Saba**: Mistral's Arabic-specialized model¹¹

6 Recommendations and Future Outlook

6.1 Leading Models for Initial Integration

Based on comprehensive benchmark analysis, **Claude 3.5 Sonnet and GPT-4/GPT-4o are recommended as the leading candidates** for initial integration into the Arabic E-learning platform.

Claude 3.5 Sonnet	GPT-4/GPT-4o
Ideal for nuanced Arabic understanding, cultural context, and complex query handling. Exceptional ADMD performance indicates deep Arabic comprehension.	Superior for broad Arabic knowledge, reasoning tasks, and educational content generation. Strong ArabicMMLU performance demonstrates comprehensive knowledge.

6.2 Continuous Evaluation Strategy

Current benchmarks have limitations¹² that necessitate ongoing evaluation with diverse Arabic datasets to ensure optimal model performance.

- Regular testing against updated Arabic benchmarks
- Custom evaluation metrics for pedagogical effectiveness
- Monitoring cultural appropriateness and accuracy
- Assessing conversational quality and grammar correction

⁹<https://www.linkedin.com/pulse/building-generative-ai-applications-langchain-sunil-kumar-u6ygc>

¹⁰<https://en.incarabia.com/core42-announces-opensource-arabic-llm-jais-30b-617603.html>

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¹²<https://aclanthology.org/2024.findings-acl.334.pdf>

6.3 Fine-tuning & Customization Potential

While leading proprietary models offer strong out-of-the-box performance, open-source Arabic LLMs like Jais and AceGPT ¹³ provide opportunities for fine-tuning and customization.

Recommendation: Consider hybrid approach using leading proprietary models for production while exploring fine-tuning of open-source models for specific educational needs.

Conclusion

This comparative analysis identifies Claude 3.5 Sonnet and GPT-4/GPT-4o as the leading LLM candidates for Arabic e-learning applications, based on their superior performance across multiple Arabic-specific benchmarks. Claude 3.5 Sonnet excels in nuanced Arabic understanding (30% ADMD accuracy), while GPT-4 demonstrates exceptional broad Arabic knowledge (72.5% ArabicMMLU). Both models offer robust LangChain integration and API support, making them suitable for immediate platform integration. Organizations should consider a hybrid approach, leveraging these proprietary models while exploring open-source alternatives like Jais and AceGPT for specialized fine-tuning opportunities.

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¹³<https://en.incarabia.com/core42-announces-opensource-arabic-llm-jais-30b-617603.html>